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Rapport of AI Internship

Done at **Forvia**

**Application de la data science et
engineering pour l'extraction intelligente de
données projets dans l'environnement de
l'entreprise**

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List of abbreviations

- **AIP** = **A**rtificial **I**ntelligence **P**latform
- **FIT** = **F**orvia **I**formatique **T**unisie
- **GIT** = **G**lobal **I**nformation **T**chnologies
- **KPI** = **K**ey **P**erformance **I**ndicator
- **SLA** = **S**ervice **L**evel **A**greement

General Introduction

In a context marked by growing digitalization and the explosion of data volumes, industrial companies seek to fully leverage their information systems in order to improve processes and optimize decision-making. Data has become a strategic asset, not only to ensure operational performance but also to foster innovation and maintain competitiveness.

However, the dispersion of data across different systems, the complexity of management tools (SAP, data platforms, process automation), and the diversity of business needs represent major challenges. Teams require reliable, consolidated, and accessible information to monitor key activities, identify bottlenecks, and anticipate risks.

It is within this framework that our Assistant Engineer Internship project, entitled “*Application de la data science et engineering pour l’extraction intelligente et la prédiction de données projets dans l’environnement de l’entreprise*”, was carried out at Forvia. The main objective of this project is to develop dynamic and interactive dashboards to visualize key performance indicators (KPIs) related to incident and approval processes, with the aim of enhancing operational monitoring and supporting informed decision-making.

To achieve this, the project followed a structured set of phases, ranging from business needs analysis and KPI identification to data preparation, dashboard implementation, and user evaluation.

This report is organized into the following 3 main chapters : The first chapter introduces the host organization and presents the tools and technologies that will be used for this project. The second chapter outlines the project’s main objectives. Finally, the third and last chapter focuses on the realisation and evaluation of our project.

GENERAL CONTEXT AND FRAMEWORK

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Introduction

In this chapter, our aim is to describe the context of our project. We will introduce the host organization, the team I will be joining, and the various tools and technologies that will be used throughout the project.

1.1 Host Organisation

This section is dedicated to the presentation of Forvia. Forvia is a global group specializing in automotive technologies, resulting from the merger between Faurecia and HELLA in 2022. Present in more than 40 countries, it brings together over 150,000 employees and positions itself as a key player in sustainable mobility. The group is structured into six international Business Groups, supported on the IT side by the transversal business group, Global Information Technologies.

Figure 1.1 shows the distribution of the various GIT teams around the world.



FIGURE 1.1 : Distribution of GIT teams around the world

Forvia Informatique Tunisie (FIT) is a key member of Forvia's transversal Business Group, Global Information Technologies. It is structured into 12 departments.

Figure 1.2 presents the various Forvia's departments.

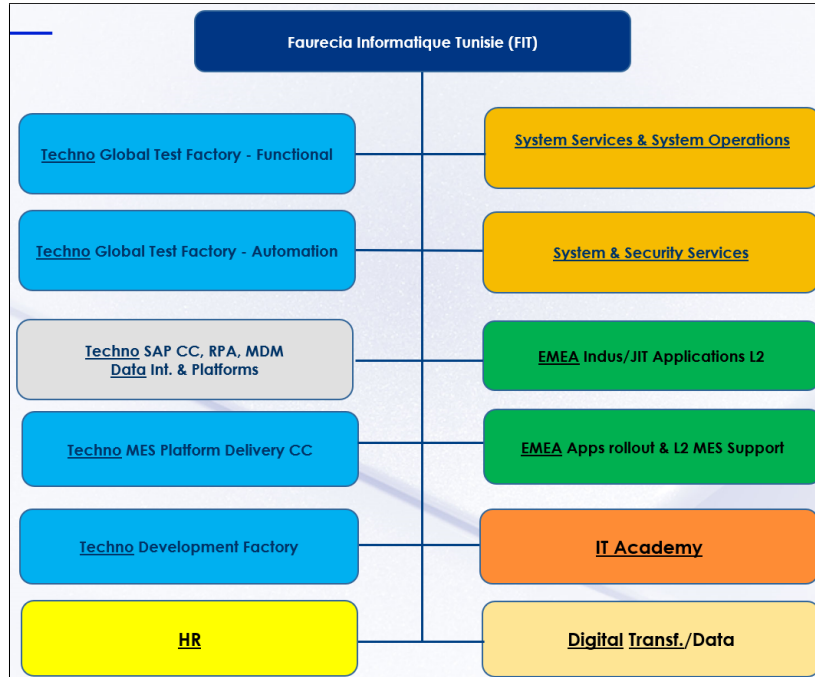


FIGURE 1.2 : Presentation of the various Forvia's departments

The **Techno SAP CC, RPA, MDM, Data Integration & Platforms** department is responsible for designing, maintaining, and optimizing Forvia's core digital backbone. It ensures efficient business operations by developing and customizing SAP solutions, automating processes through RPA, centralizing and harmonizing master data with MDM, and enabling seamless data integration and ingestion via platforms such as Palantir and Informatica.

It is in this department that this internship will take place, more specifically in the Data Integration team. This team manages data ingestion pipelines and integration flows from multiple sources, such as Palantir and Informatica, to provide unified and trusted data solutions to other teams. The team has two main activities, each one with its own work process :

- **Do Projects** : This activity consists of carrying out Palantir and Informatica projects, either directly on the respective applications or based on requests received by mail. The work follows a DIA, which represents the client's guidelines and expectations for the task to be completed.
- **Ticket Resolution** : This activity consists of solving tickets on the company's ticketing platform. There are two types of tickets :
 - **Incident tickets** : Related to unexpected issues, errors, or malfunctions that need corrective actions to restore normal operations.
 - **Approval tickets** : Requests that require validation or authorization from specific stakeholders before being processed further.

1.2 Tools and technologies review

1.2.1 Palantir

We will start first by talking about Palantir. Palantir is the tool used by Forvia to manage data. The graph 1.3 shows the different layers that compose this tool [1].

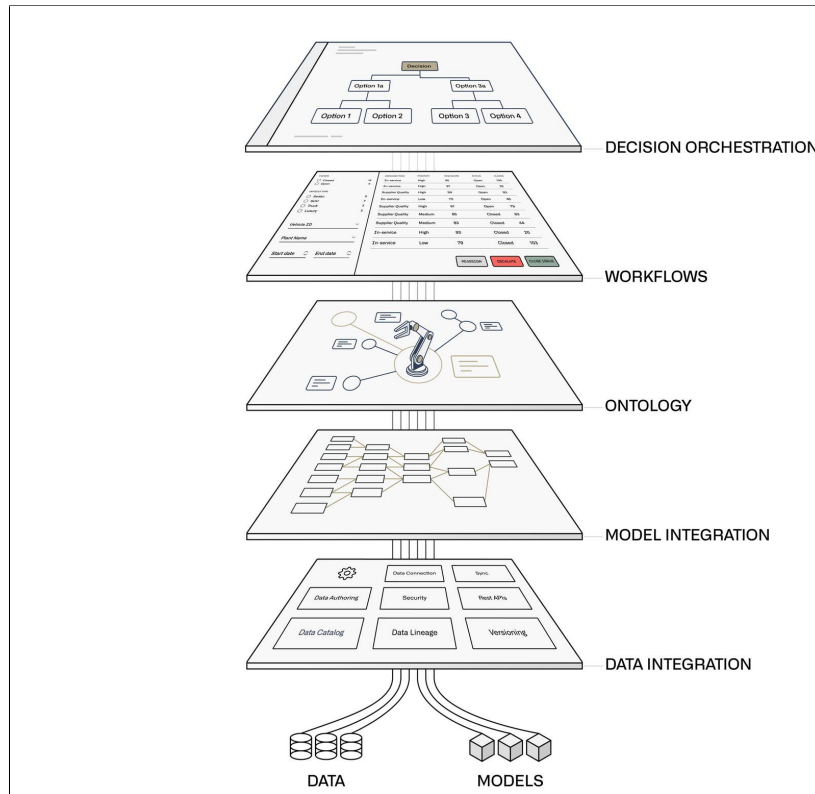


FIGURE 1.3 : Palantir's different layers

Palantir Foundry offers a modular architecture that transforms raw data into actionable decisions. This structure, made up of several functional layers, enables smooth, governed, and collaborative data operations from ingestion to execution as follows :

- **Data Integration** : Connects data sources and ensures security, traceability, and version control. It provides the technical foundation for robust data governance.
- **Model Integration** : Allows integration of analytical or machine learning models to enrich data or support decision-making.
- **Ontology** : Translates technical data into business objects (e.g., Machine, Ticket), facilitating analysis and cross-functional collaboration.
- **Workflows** : Provides interactive interfaces to manipulate business objects, monitor processes,

and take action on data.

- **Decision Orchestration** : Orchestrates strategic decisions based on data, rules, or models, either in guided or automated modes.

1.2.2 Service Now

ServiceNow is a powerful cloud platform designed to streamline the way organizations handle their internal services and operations. It helps centralize requests, incidents, changes, and tasks through structured processes, reducing manual operations and improving service delivery [2].

Figure 1.4 presents ServiceNow's workflow.



FIGURE 1.4 : Palantir's different layers

The diagram shows how key ITSM processes in ServiceNow—like Incident, Problem, Change, Request, and the Service Catalog—all revolve around the Service Desk. These processes are connected and supported by the CMDB, ensuring smooth, reliable service delivery.

Each of these processes plays a distinct role in the overall ITSM strategy, as summarized below :

- **Data Integration** : Simplifies logging, categorization, SLA monitoring, and restoring disrupted services rapidly
- **Problem Management** : Focuses on root cause analysis to prevent recurrent issues by addressing systemic faults, beyond just incident resolution
- **Change Management** : Automates change approval workflows, minimizes risk through structured flows (normal, emergency, dev-ops), and ensures governance across releases.

- **Service Request & Catalog** : Empowers users via self-service portals to request services or assets, reducing manual dispatch and improving speed .
- **CMDB** : Maintains a comprehensive asset and configuration repository that maps relationships and supports incident understanding, problem triage, and impact analysis.

Conclusion

In this chapter, we provided a brief introduction to the host organization and presented the various tools and technologies that will be used in the project. In the next chapter, we will outline the project details and explain how we intend to approach it

PROJECT OBJECTIVES

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Introduction

In this chapter, we aim to review the project details and explain how we intend to approach it. We will begin by presenting the review and analysis of the existing system, the problematic and needs study followed by our work methodology, the project life cycle, and finally the Gantt chart.

2.1 Review and analysis of the existing system

Project monitoring within the Data Integration team follows the process below :

- **List the team's tickets** : Search for the team's tickets in ServiceNow.
- **Identify delayed tickets** : Calculate the duration between the creation date and the closure date, then compare it to the SLA target. Keep only the tickets that exceed the SLA.
- **Retrieve information on delayed tickets** : Select the delayed tickets and gather relevant information from various ServiceNow tables.
- **Annotate the most recurring problems** : List all identified problems and their frequency in order to determine the most recurring ones.
- **Manually create charts** : Create charts manually to illustrate the recurring problems.

2.2 Problematic

At first glance, the actual project monitoring may seem simple. However, in reality, it involves several limitations that hinder both the project leader and the team members.

- **Time-Consuming Processes** : The absence of automation in key steps such as ticket analysis, data retrieval, and chart creation leads to repetitive manual work that slows down project monitoring and reduces overall efficiency.
- **Limited Visibility** : The system does not provide a centralized or real-time view of the project status, making it difficult for the project leader to track progress efficiently.
- **Scattered Information** : Project data is stored across multiple ServiceNow tables or systems, requiring users to manually cross-check and consolidate information.
- **Difficult KPI Tracking** : Key performance indicators are not easily extractable or visualized.
- **Lack of Adaptability** : The monitoring system is rigid, and hard to adapt to specific project needs, new KPIs, or changes in workflow

2.3 Proposed solution and project's key objectives

In response to these challenges, we propose to develop a decision-making dashboard for monitoring both approval and incident tickets.

To achieve this, the following steps are planned :

- Prepare and define the key points of the project ;
- Design the project from both a business and technical perspective ;
- Collect the relevant information from the ServiceNow tables ;
- Transform and clean the data ;
- Build and configure the decision-making dashboard.

2.4 Needs analysis

2.4.1 Functional requirements analysis

- **Preparation** : The first phase consists in defining the key elements of the project, such as its objectives, scope, team, governance, budget, expected benefits, and timeline. Then, a study of the existing system is carried out to better identify its weaknesses and limitations.
- **Conception** : The second phase focuses on identifying the business and technical requirements of the project. This step helps define the system's key features and functionalities.
- **Collection** : The third step focuses on collecting the necessary data from the various ServiceNow tables.
- **Transforming** : The fourth phase consists in transforming the raw data collected from the tables into usable data by performing operations such as table joins, normalization, and data cleaning.
- **Building** : The final phase aims to build and configure a decision-making dashboard that summarizes all the key elements for project monitoring.

2.4.2 Non-functional requirements analysis

In addition to the project's functional specifications, an analysis of non-functional requirements is essential for the successful execution of the project.

Therefore, we have defined two major non-functional needs that we aim to meet :

- **Decision-making time efficiency** : Given that the market is becoming increasingly competitive and that acquiring new projects depends on the quality and proper monitoring of completed ones, the efficiency to time ratio has become a key factor in maximizing company profits by enabling faster and more accurate decision-making.
- **Usability** : Adding to the efficiency, we must ensure hhat the dashboard is intuitive and easy to use, allowing users to access insights with minimal training.

2.5 Project lifecycle

To carry out the project in a structured and efficient way, we defined a clear project lifecycle that breaks the work into distinct, organized phases. For this purpose, we divide our project into 6 essential phases :

- **KPI Identification** : Understand from the business which aspects they want to monitor in order to create relevant KPIs.
- **Data Comprehension and Connection** : Understand the data from a business perspective and identify the corresponding technical elements. Then, upload the relevant data identified in the previous phase into the Palantir environment.
- **Data Transformation** : Process and clean the data through operations such as casting, joining, filtering, and dropping irrelevant fields.
- **Data Analysis** : Display all the created and generated insights in a decision-making dashboard.

2.6 Gantt Chart

Finally, to ensure the smooth progress of the project and adherence to the final deadline, we defined a project schedule with specific deadlines for each phase.

The figure 2.1 provides an overview of the project's Gantt chart.

	W1	W2	W3	W4	W5	W6	W7	W8
Trainings								
Preparation								
KPI Identification								
Data Comprehension and Connection								
Data Transformation								
Data Analysis								

FIGURE 2.1 : Project Gantt Chart

The Gantt chart shown above illustrates the project timeline, dividing its execution into several phases and tasks, each with a defined duration and planned start and end dates.

The first week will be dedicated to training on the tools required for this project. During the second week, we will focus on the preparation phase. The following two weeks will be allocated to KPI identification. In the fifth week and the first half of the sixth week, the focus will be on data comprehension and establishing connections. The remaining part of the sixth week, along with the seventh week, will be dedicated to data transformation. Finally, the last week will be reserved for the data analysis phase.

Conclusion

In this chapter, we reviewed and analysed the existing system, outlined its main challenges, and presented the proposed solution along with the project's key objectives. We also described the chosen approach and defined the corresponding deadlines. In the next chapter, we will provide a detailed presentation of the project's implementation as well as its evaluation.

REALISATION AND EVALUATION

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Introduction

In this chapter, we are going to present how our project was implemented and evaluated. We will begin by covering all the steps, from identifying the relevant KPIs to building the final dashboard.

3.1 KPI Identification

The first step in the creation of our project was to understand what the stakeholders needed to evaluate and visualize in order to extract the relevant KPIs for our dashboard.

After several meetings with the business stakeholders, we identified and listed the following KPIs :

For the monitoring of incidents, the chosen KPIs are :

- **Number of open tickets** : Measures the total incidents still unresolved at a given time.
- **Percentage of closed tickets** : Tracks the share of incidents that have been successfully resolved compared to the total.
- **Average response time** : Calculates the mean time taken to respond to an incident from its creation.
- **Workload** : Shows the volume of incidents created and closed over a selected period.
- **State of the tickets** : Indicates the current status of incidents (e.g., new, in progress, resolved, closed).
- **Urgency of the tickets** : Reflects the criticality of incidents, helping prioritize resolution.
- **Number of ticket reassignments** : Counts how many times a ticket was transferred between different teams or agents.
- **Number of reopened tickets** : Measures how many incidents were marked as resolved but reopened for further action.
- **Tickets provenance** : Identifies the source or origin of incidents, providing context for trend analysis.

For the monitoring of approvals, the chosen KPIs are :

- **Total tickets on delay** : Counts the number of approval tickets that were not completed within the expected timeframe.

- **Delay rate** : Measures the percentage of approval tickets that experienced delays compared to the total number of tickets.
- **Average response time** : Calculates the mean time taken to complete an approval from its creation.
- **Workload** : Shows the volume of approval tickets created over a selected period.
- **SLA compliance** : Indicates the proportion of tickets resolved within the Service Level Agreement, reflecting process efficiency.
- **Done by robot or not** : Tracks whether an approval was completed automatically by a system/robot or manually by a human.

3.2 Data Comprehension and Connection

To achieve these KPIs, we must first understand the data at our disposal.

ServiceNow includes over 3000 tables that store and manage a wide variety of data across the platform [3]. These tables cover everything from IT service management to business operations. Instead of trying to work with all the available data, it's more practical to focus only on the tables that are truly relevant to our analysis.

That's why we begin by deeply exploring the two main tables : *Incident* and *Approval*.

The *Incident* table contains 126 columns that capture the complete lifecycle and context of an incident :

- **Identifiers** (sys_id, number) : uniquely identify each incident and enable tracking across the platform.
- **Timestamps** (opened_at, closed_at, resolved_at) : record key dates in the incident lifecycle.
- **Durations** (business_duration, calendar_duration) : measure how long an incident remained open, in both business and calendar time.
- **Priority and impact** (priority, urgency, impact, severity) : indicate the criticality of the incident to guide resolution priorities.
- **Assignment data** (assignment_group, assigned_to, reassignment_count) : track which users or teams were responsible for handling the incident.
- **Business context** (business_service, cmdb_ci, company, u_application_service, u_site_name, location) : link the incident to related services, infrastructure, and organizational units.

- **SLA tracking** (sla_due, made_sla) : determine whether the service level agreements were met.
- **Custom fields** (u_*) : provide additional domain specific information such as alert origin, escalation status, and customer impact.
- **Categorization** (category, subcategory, cause, u_root_cause) : classify the incident and identify its cause.
- **Workflow tracking** (state, incident_state, approval, close_code) : monitor the incident's progress and closure.

From this analysis, we conclude that the following additional tables are needed alongside the *Incident* table :

- **Group** table : provides details about "Assignment data" specifically the assignment groups, ensuring we correctly identify and track incidents assigned to our teams.
- **Location** table : enriches the *Business context* by providing detailed information about the geographic or organizational origin of each ticket.

The *Approval* table contains 35 columns that describe the approval process of ServiceNow records. It includes fields related to the approver, the approval workflow, and the status of each approval request :

- **Identifiers** (sys_id, document_id, u_number) : uniquely identify the approval record and the related document.
- **Approval source** (approval_source, source_table, sysapproval, sysapproval_link) : define which record triggered the approval and from which table it originated.
- **Approver info** (approver, approver_link, group, group_link) : capture the individual or group responsible for approving the request.
- **Approval status** (state, state_binding, state_binding_link) : track the current status of the approval (e.g., requested, approved, rejected).
- **Workflow integration** (process_step, process_step_link, wf_activity, wf_activity_link) : link the approval to its place within a workflow or process.
- **Dates** (due_date, expected_start, sys_created_on, sys_updated_on) : provide timing information on when the approval was created, is expected to start, and is due.

- **Metadata** (sys_created_by, sys_updated_by, sys_mod_count, sys_tags) : capture authorship, update history, modification count, and tagging.
- **Comments** (comments, approval_journal_column) : store any notes or justifications related to the approval process.
- **Custom fields** (u_approval_advice, u_business_group) : include additional business specific input or categorization.

Based on this analysis, we conclude that the following additional tables are needed alongside the *Approval* table to provide complete context and enrich our analysis :

- **Group** table : provides details about assignment groups, helping us identify and track approvals related to specific teams.
- **sc_req_item** table : links approvals to service catalog requests, enabling us to trace approvals back to specific catalog items.
- **sysapproval_group** table : stores grouped approval records and helps track the overall approval flow associated with a given request or task.
- **cmdb_ci_service** table : provides information about business or technical services tied to the approved item, useful for understanding service level impact.
- **sys_user_grmember** table : defines relationships between users and groups, allowing us to determine group membership of approvers.
- **sys_user** table : stores user profiles and details needed to understand who the approvers are (roles, departments, locations, etc.).
- **Mass UploadActivity Report** table : It's an excel table that is manually filled by our team to track information that is not available in Service Now such as if the approval was done by a robot or a humain.

Finally, after identifying all the needed tables from Service Now we are going to upload them on Palantir using the "Data Connection" function. In the case of Service Now tables, there is a specific API connector that allows us to import all the data easily and synchronize its update automatically each week. For the excel table, we will use a specific API connector with Sharepoint allowing us also to import all the data easily and synchronize its update automatically each week.

3.3 Data Transformation

Once all the required tables have been uploaded, we proceed with a transformation and cleaning process to prepare the data for analysis :

- **Initial Cleaning** : This step involves dropping all unnecessary or completely empty columns before performing any other operations. It helps reduce processing time and resource usage by working only with relevant data.
- **Joining** : We combine information from multiple tables using common keys (columns containing the same values), enriching the dataset with related attributes.
- **Feature Engineering** : New features are created by applying operations and transformations on existing data to extract more meaningful or aggregated insights.
- **Final Cleaning** : This includes casting data types, renaming fields if necessary, and removing any remaining irrelevant information—retaining only what is required for the KPIs.

Let's now have a closer look at the specific process done on each of the incidents and approvals tables.

The figure 3.1 illustrates the process of cleaning and transformation of the incidents table.

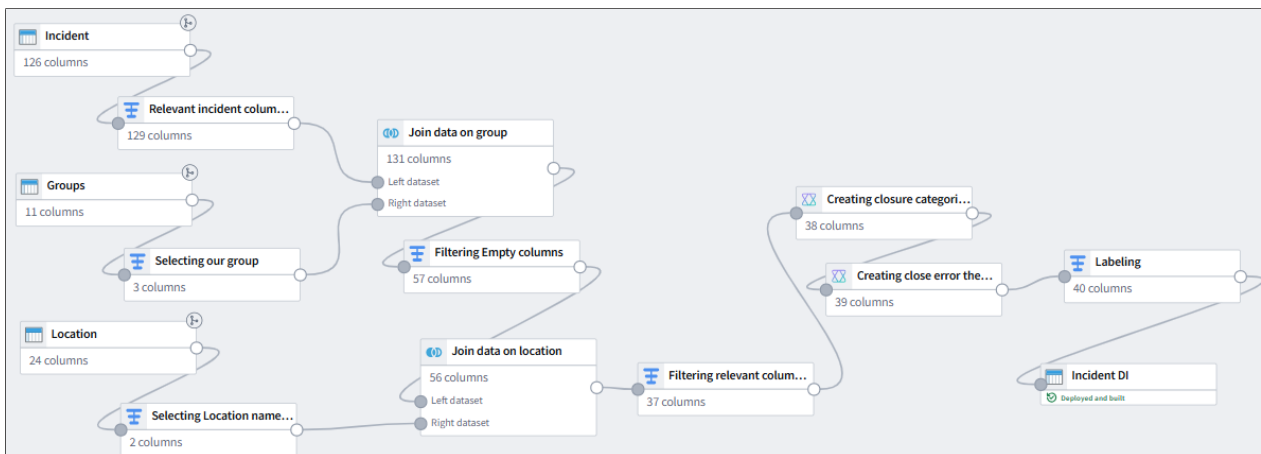


FIGURE 3.1 : Process of cleaning and transformation of the incidents table

As we can see in the previous figure, we followed the general transformation and cleaning process mentioned earlier. First, we cleaned all unnecessary columns from the uploaded tables. For example, in the *Incident* table, we dropped columns unrelated to the KPIs; in the *Groups* table, we selected only our relevant group; and in the *Location* table, we retained only the necessary columns. Next, we joined the tables and removed any empty columns resulting from each join.

Afterwards, we used the **Artificial Intelligence Platform (AIP)** to create new semantic features. Specifically, for each ticket's close note, we categorized it into five predefined classes. Before applying this to our data, we tested AIP on a labeled enterprise dataset similar to ours. Starting with a simple prompt, we achieved 87% accuracy, and after refining the prompt, we reached 91% accuracy, which is satisfactory for our task.

We then used AIP again to extract specific fields from text, this time to identify the closure theme from close notes. Unlike the previous step, we did not have a labeled dataset, so we cannot report a statistically valid precision. However, based on a manual review of a substantial sample, all extracted themes appeared correct.

Finally, we cast columns with mismatched types, converted business and calendar durations to hours, and added semantic meaning to numerical variables, for example, replacing the value 1 in *state* with New and similar transformations.

The figure 3.2 illustrates the process of cleaning and transformation of the approval table.

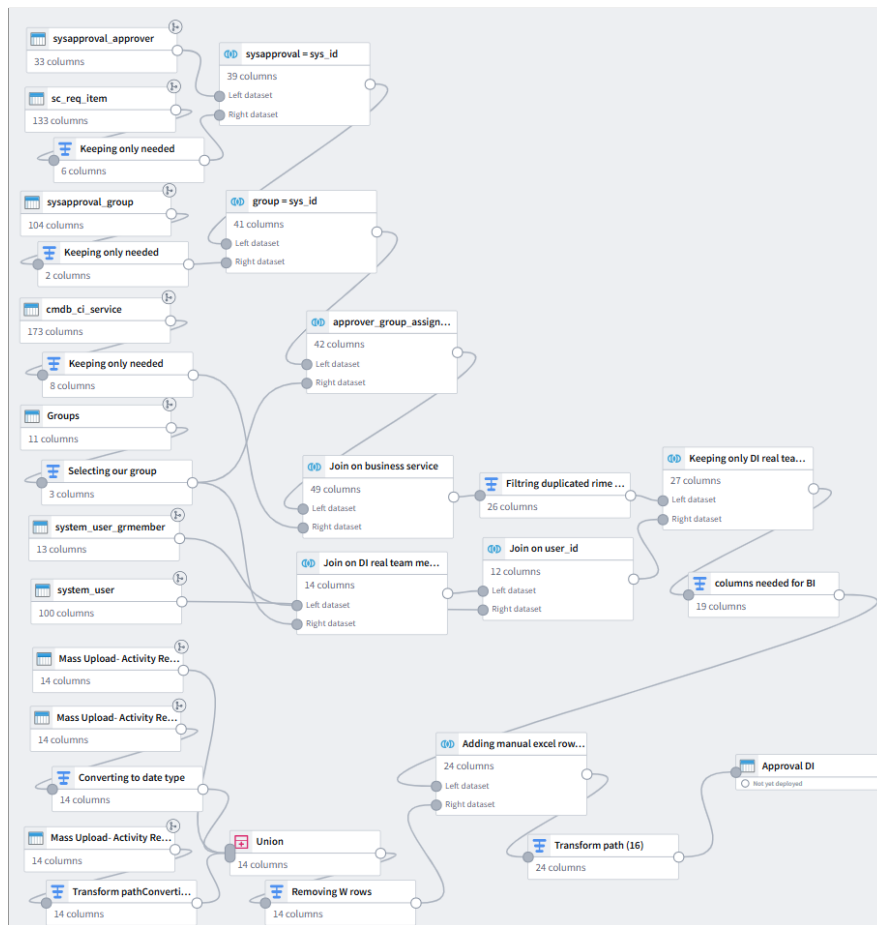


FIGURE 3.2 : Process of cleaning and transformation of the approvals table

As done before, we followed the general transformation and cleaning process. We started by

eliminating unneeded columns and keeping only the relevant ones. After that, we moved on to joining the tables. This time, we added an intermediate step : performing a union to merge the three Excel files containing the Mass Upload Activity Reports for the years 2023, 2024, and 2025.

Afterwards, we created new semantic features by manipulating other columns. For instance, we introduced the *effective duration* column, which tracks how many business hours it takes to complete an approval by subtracting the creation date from the last update date. We also added a *same day* column to indicate whether the ticket was closed on the same day it was created, and a *less than 24 hours* column to mark tickets resolved within a single day.

Finally, as with the incident data, we cast columns with mismatched types and enriched numerical variables with semantic meaning like replacing the value 1 in *urgency* with “Critical,” along with similar transformations.

Now, we have both the incident and approval datasets cleaned and ready to be used in the dashboard for analysis.

3.4 Data Analysis

With the new cleaned datasets, we are now ready for the creation of our dashboards.

Figure 3.3, 3.4 and 3.5 give a perspective of the incident dashboard.

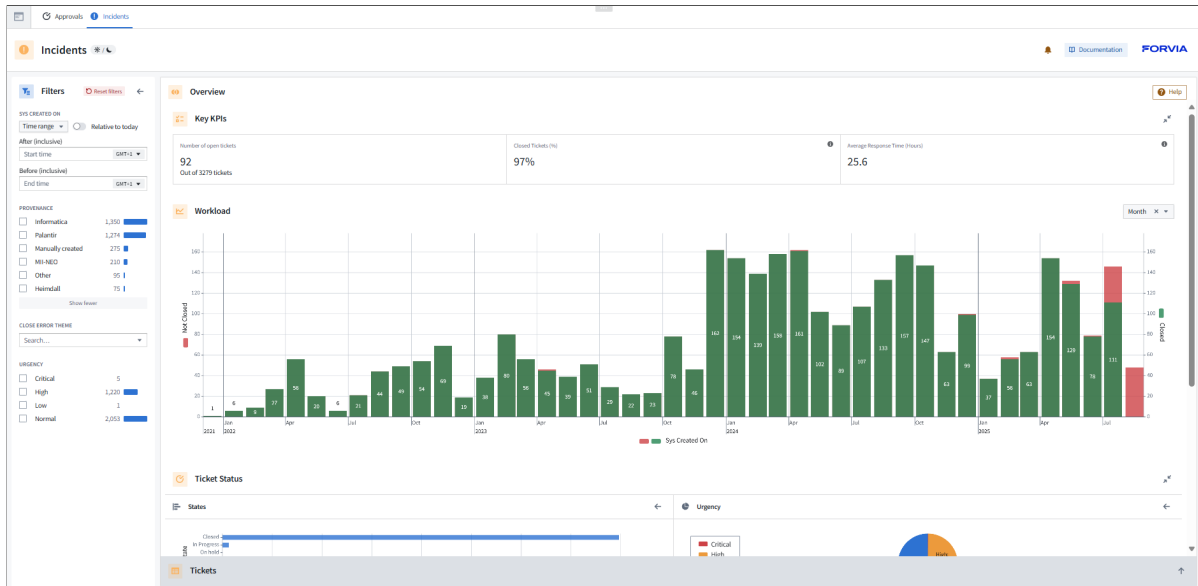


FIGURE 3.3 : Incident dashboard part 1

As the figure shows, our dashboard is divided into 3 main parts :

- **Toolbar** : Contains the navigation between the pages *Incident* and *Approval*, a *Documentation*

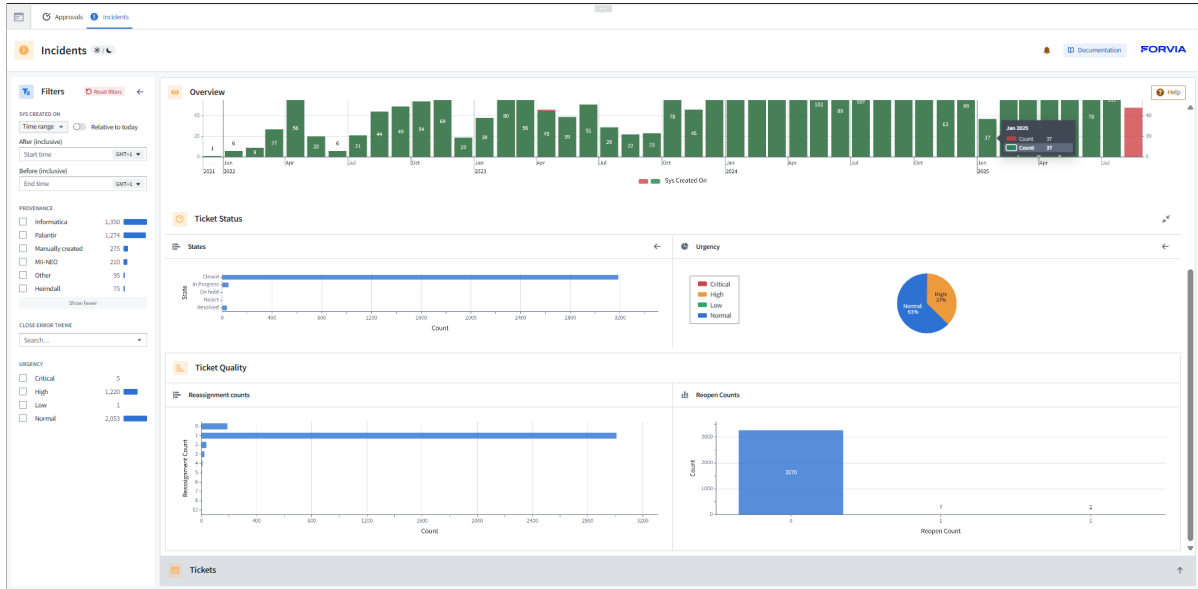


FIGURE 3.4 : Incident dashboard part 2

button linking to the Palantir documentation, a dark/light mode toggle, and a notification button that alerts the user once the dashboard is updated.

- **Filters** : The central element that directly filters the *Overview* part. It contains multiple sub-filters to refine the displayed information, as well as a *Reset Filters* button to clear all active selections.
- **Overview** : Displays all graphs, metrics, and decision elements organized in different sections. The graphs in this part also act as interactive filters for the other sections, from top to bottom. It also includes a *Help* button that links to the internal chatbot for additional assistance.

It is important to mention that every part, section, and subsection is collapsible, allowing the user to focus on specific elements by hiding those that are not relevant at a given moment.

Let us first take a closer look at the *Overview* part, which is divided into 5 major sections :

- **Workload** : A bar chart showing the number of incidents created and closed per week, month, or year. This allows us to easily identify periods with unresolved incidents. Clicking on a period filters both the *Key KPIs* and *Ticket Status* sections.
- **Key KPIs** : Displays key metrics that should be immediately visible when opening the dashboard. In the case of incidents, this metrics are the number of tickets still open as of the latest update, the percentage of closed tickets, and the average response time. These values update dynamically when a period is selected in the workload chart.

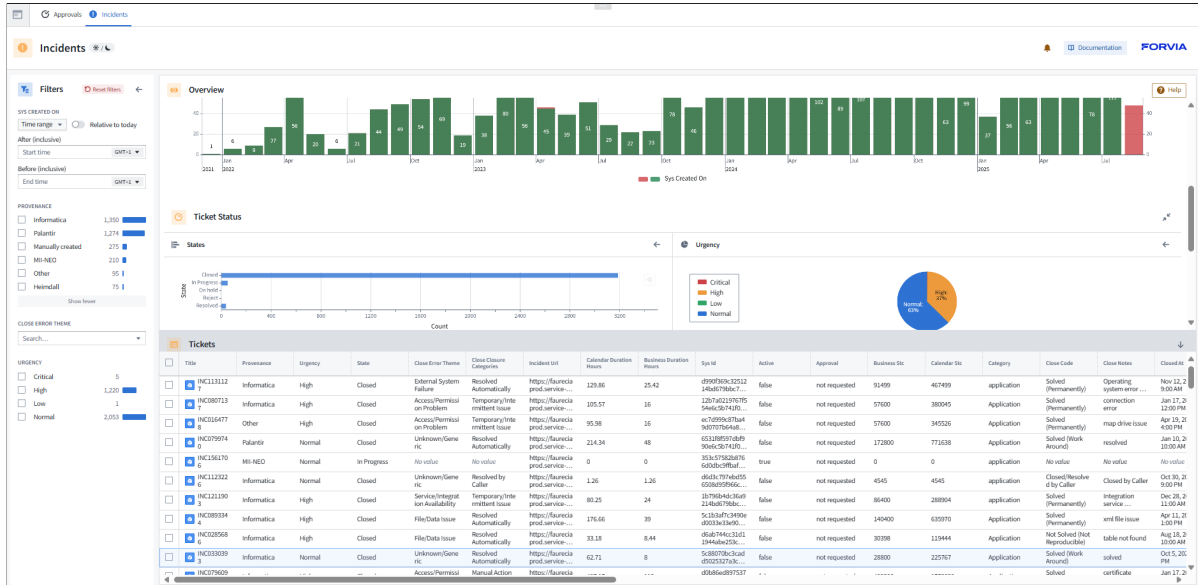


FIGURE 3.5 : Incident dashboard part 3

- **Ticket Status** : Tracks ticket states and their urgency. Selecting a state reveals the urgency distribution of its tickets.
- **Ticket Quality** : Shows the number of reassignments for tickets selected in the urgency view of the *Ticket Status* section.
- **Tickets** : Lists all tickets. Clicking on one displays its full details, making it easier to identify and investigate problematic tickets.

Moving now to the *Filters* part, which is divided into four sections :

- **Sys Created On** : Two calendars to select the start date and end date.
- **Provenance** : A checklist containing the different sources from which incidents originate.
- **Close Error Theme** : A scrollable list of all closing error themes.
- **Urgency** : A checklist with the urgency levels of incidents. This filter is included even though urgency already appears in the chart, as the pie chart for urgency depends on the ticket state. Here, the checklist provides an independent view of urgency.

Figure 3.6 and 3.7 give a perspective of the approval dashboard.

As the figure shows, our dashboard is also divided into 3 parts identically to the Incident dashboard.

First, we consider the *Overview* part, which is divided this time into 4 major sections :

- **Workload** : A bar chart showing the number of approvals created per week, month, or year. This allows us to easily identify periods of peak and charge. Clicking on a period filters both the *Key KPIs* and *Ticket Quality* sections.

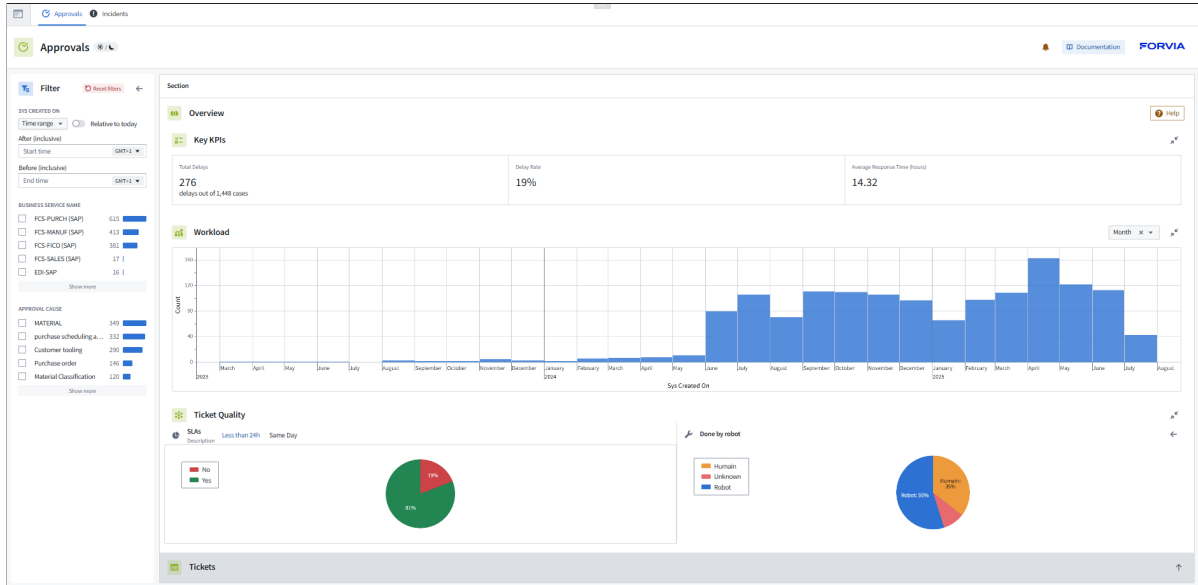


FIGURE 3.6 : Approval dashboard part 1

- **Key KPIs** : Displays key metrics that should be immediately visible when opening the dashboard. In this case, the key metrics are the number of total delays as of the latest update, the percentage of delay rate, and the average response time in hours. These values update dynamically when a period is selected in the workload chart.
- **Ticket Quality** : Shows the respect of the SLA and if the approval is done automatically by a robot or manually by a human. The respect of SLA subsection shows the percentage of the tickets resolved in less than 24 hours and the percentage of the tickets resolved the same day.
- **Tickets** : Lists all tickets. Clicking on one displays its full details, making it easier to identify and investigate problematic tickets once the filters are applied.

Next, we consider the *Filters* part, which is divided into 3 sections :

- **Sys Created On** : Two calendars to select the start date and end date.
- **Business Service Name** : A checklist containing the different Business services from which incidents originate.
- **Approval Cause** : A checklist of the main approval causes.

During the creation of the dashboards, an evaluation phase was carried out to ensure their quality and usefulness. The evaluation was conducted along several dimensions, such as clarity, ease of navigation, and overall user experience.

To perform this evaluation, we organized feedback sessions with the main target users of

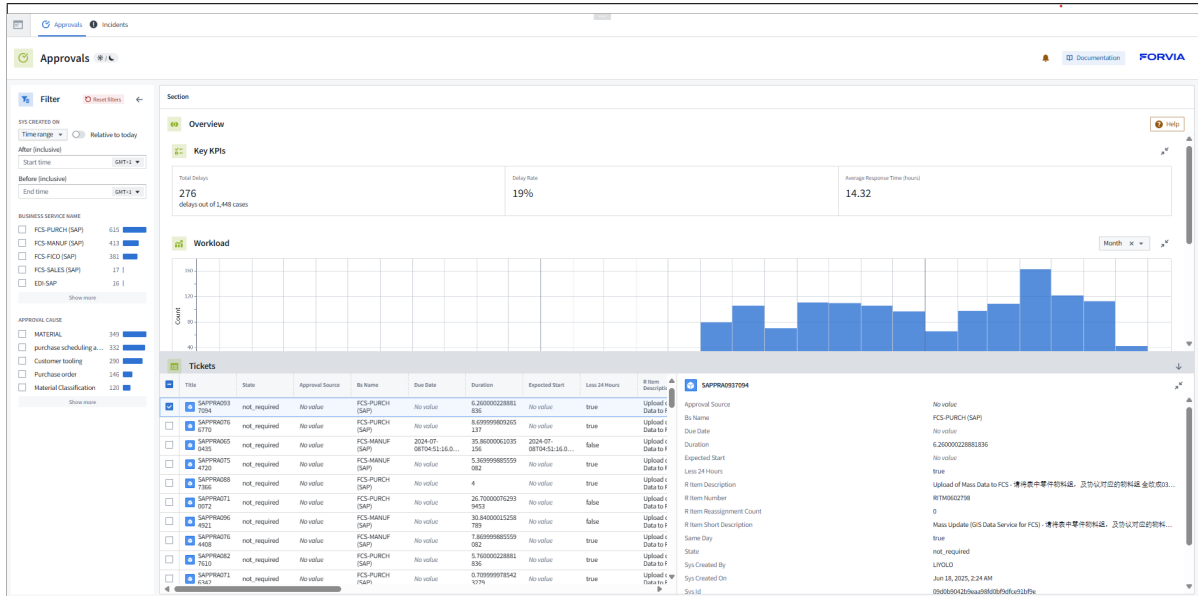


FIGURE 3.7 : Approval dashboard part 2

these dashboards. During these sessions, participants tested the dashboards in real conditions and provided feedback regarding :

- **Readability and Clarity** : Whether the KPIs, charts, and filters were easy to understand and interpret at first glance.
- **Relevance** : To what extent the dashboards answered their operational and strategic questions.
- **Interactivity** : The effectiveness of filters, drill-down options, and collapsible sections in allowing smooth exploration.
- **Performance** : Responsiveness and refresh times when interacting with the dashboard.
- **Actionability** : Whether the dashboards provided actionable insights for decision making.

Based on this feedback, several iterative adjustments were made to improve the usability of the dashboards such as reorganizing sections, refining certain KPIs, and clarifying labels. This user-centred evaluation ensured that the final dashboards, that we have seen above, were not only technically functional but also aligned with the needs of the business stakeholders.

Conclusion

In this chapter, we identified the KPIs to be used, analyzed the data in order to import only the relevant ones, performed various cleaning and transformation steps, and finally concluded with the creation of two decision-making dashboards.

General Conclusion and Perspectives

Throughout this project, we focused on the development of interactive dashboards to monitor incidents and approvals within ServiceNow, providing actionable insights for the business. We extracted and imported data from multiple tables, including Incident, Approval, and supplementary sources, and performed extensive cleaning, transformation, and feature engineering to prepare the datasets for analysis.

During this work, we faced several challenges, notably understanding the complexity of the data structure and ensuring the accuracy of derived KPIs. However, by iteratively refining our approach and using AI-assisted tools for semantic analysis, we were able to create dashboards that are both reliable and user-friendly.

As a next step, the project could be enhanced by integrating a machine learning process capable of predicting the expected duration to resolve a ticket upon its creation. This addition would allow for better planning of workloads, proactive management of SLA compliance, and more informed allocation of resources, making the dashboards even more actionable for decision-making.

This project allowed me to grow technically by exploring data integration, transformation, and visualization for the first time at this scale. I also developed personally by working autonomously, setting ambitious goals, and improving my time and project management skills.

Bibliographie

- [1] P. TEAM. « About palantir. » [Last Access on the 01-07-2025]. (12 Avril 2024), adresse : <https://www.palantir.com/about/>.
- [2] S. N. TEAM. « What is ServiceNow? » [Last Access on the 02-07-2025]. (), adresse : <https://www.servicenow.com/what-is-servicenow.html>.
- [3] S. N. TEAM. « Create the NeedIt Application and Application Files. » [Last Access on the 06-07-2025]. (), adresse : https://developer.servicenow.com/dev.do#!/learn/learning-plans/xanadu/new_to_servicenow/app_store_learnv2_buildneedit_xanadu_tables.